# Mathematics II

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You can find the last version of these course materials at

https://mbujosab.github.io/MatematicasII/

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## 1 Highlights of **Lesson 4**

### Highlights of Lesson 4

- Elementary transformations
- Identifying singular matrices by elimination
- Matrix multiplication of elementary matrices





## 2 Elementary transformations of a matrix

Type *I*: $\mathbf{A}_{\overset{\tau}{[(\lambda)i+j]}}$ (with $i \neq j$)

add $\lambda$ times $i$-th column ($\lambda\mathbf{A}_{|i}$) to $j$-th column ($\mathbf{A}_{|j}$)

$$\begin{bmatrix} 1 & -3 & 0 \\ 1 & -6 & 3 \end{bmatrix}_{\overset{\tau}{[(-2)\mathbf{1}+\mathbf{3}]}} = \begin{bmatrix} 1 & -3 & -2 \\ 1 & -6 & 1 \end{bmatrix}$$

Type *II*: $\mathbf{A}_{\overset{\tau}{[(\alpha)i]}}$ (with $\alpha \neq 0$)

multiply by $\alpha$ the $i$-th column

$$\begin{bmatrix} 1 & -3 & 0 \\ 1 & -6 & 3 \end{bmatrix}_{\overset{\tau}{[(10)\mathbf{2}]}} = \begin{bmatrix} 1 & -30 & 0 \\ 1 & -60 & 3 \end{bmatrix}$$

## 3 Elimination and pre-echelon form of a matrix

- *Pivot* is the first non-zero component of each column.
- *Elimination*: modifies a matrix until all components at the right-hand side of each pivot are zeros

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 0 \\ 2 & 8 & 4 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{\tau_{[(-3)\mathbf{1}+\mathbf{2}]}} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 4 \\ 1 & -2 & 1 \end{bmatrix} \xrightarrow{\tau_{[(-2)\mathbf{2}+\mathbf{3}]}} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 1 & -2 & 5 \end{bmatrix} = \mathbf{L}$$





## 4 Elimination

### Elimination algorithm on **A**

modifies **A** using a sequence of *elementary transformations*

### Goal

to get a *(pre)echelon* form

- *pre-echelon*: all components on the right side of each pivot are zero.
- *echelon*: if any column before a non-null column $\mathbf{A}_{|j}$ is non-null and its pivot is above the pivot of $\mathbf{A}_{|j}$.

It is always possible to find a (pre)echelon form by elimination

**Rank** $(\mathrm{rg})$: the number of pivots in any of its pre-echelon forms

$\underset{n\times n}{\mathbf{A}}$ is *singular* if its pre-echelon forms have null-columns $(\mathrm{rg} < n)$





## 5 Elimination: When can't we find $n$ pivots?

$n \times n$ matrices are **singular** if less than $n$ pivots after elimination

$$\begin{bmatrix} 0 & 1 & 3 \\ 4 & 2 & 8 \\ 1 & 1 & 1 \end{bmatrix}$$

Has this matrix $n$ pivots? $\begin{bmatrix} 1 & 3 & 0 \\ 2 & 6 & 4 \\ 1 & 1 & 1 \end{bmatrix}$

and this one? $\begin{bmatrix} 1 & 3 & 0 \\ 2 & 6 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

and this one? $\begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & -4 \end{bmatrix}$





## 6 Matrix multiplication: elementary matrices

$$\underbrace{\begin{bmatrix} 1 & 3 & 0 \\ 2 & 8 & 4 \\ 1 & 1 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \quad & \quad & \quad \\ & & \\ & & \end{bmatrix}}_{\mathbf{I}_\tau} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 4 \\ 1 & -2 & 1 \end{bmatrix}}_{\mathbf{A}_\tau}$$

We call $\mathbf{I}_\tau$ "Elementary matrix":

$$\mathbf{A}(\mathbf{I}_\tau) = \mathbf{A}_\tau$$

This specific elementary matrix $\mathbf{I}_\tau$ is written as $\mathbf{I}_{\tau_{[(-3)\mathbf{1}+\mathbf{2}]}}$

$$\mathbf{A}\left(\mathbf{I}_{\tau_{[(-3)\mathbf{1}+\mathbf{2}]}}\right) = \mathbf{A}_{\tau_{[(-3)\mathbf{1}+\mathbf{2}]}}$$

## 7 Matrix multiplication: elementary matrices

$$\begin{bmatrix}1 & 0 & 0\\ 2 & 2 & 4\\ 1 & -2 & 1\end{bmatrix}\begin{bmatrix}\quad & \quad & \quad\\ & & \\ & & \end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 2 & 2 & 0\\ 1 & -2 & 5\end{bmatrix}$$

This specific elementary matrix $\mathbf{I}_\tau$ is written as $\mathbf{I}_{\tau_{[(-2)2+3]}}$

$$\mathbf{A}\left(\mathbf{I}_{\tau_{[(-2)2+3]}}\right) = \mathbf{A}_{\tau_{[(-2)2+3]}}$$





## 8 Elimination by elementary matrices

$$\mathbf{A} = \begin{bmatrix}1 & 3 & 0\\ 2 & 8 & 4\\ 1 & 1 & 1\end{bmatrix} \xrightarrow{\tau_{[(-3)1+2]}} \begin{bmatrix}1 & 0 & 0\\ 2 & 2 & 4\\ 1 & -2 & 1\end{bmatrix} \xrightarrow{\tau_{[(-2)2+3]}} \begin{bmatrix}1 & 0 & 0\\ 2 & 2 & 0\\ 1 & -2 & 5\end{bmatrix} = \mathbf{L}$$

$$\mathbf{A}_{\substack{\tau_{[(-3)1+2]}\\ \tau_{[(-2)2+3]}}} = \mathbf{A}_{\tau_{[(-3)1+2]}\tau_{[(-2)2+3]}} = \left(\mathbf{A}\left(\mathbf{I}_{\tau_{[(-3)1+2]}}\right)\right)\left(\mathbf{I}_{\tau_{[(-2)2+3]}}\right) = \mathbf{L}$$

there is a matrix that does the whole job "at once"

$$\mathbf{A}_{\substack{\tau_{[(-3)1+2]}\\ \tau_{[(-2)2+3]}}} = \mathbf{A}\left(\left(\mathbf{I}_{\tau_{[(-3)1+2]}}\right)\left(\mathbf{I}_{\tau_{[(-2)2+3]}}\right)\right) = \mathbf{A}\mathbf{I}_{\substack{\tau_{[(-3)1+2]}\\ \tau_{[(-2)2+3]}}} = \mathbf{L}$$

$$\boxed{\mathbf{A}_{\tau_1\cdots\tau_k} = \mathbf{A}\left(\mathbf{I}_{\tau_1\cdots\tau_k}\right)}$$





## 9 How do I get from **L** back to **A**? Inverses

How do I reverse the first step? (it was subtract 3 times $\mathbf{A}_{|1}$ from $\mathbf{A}_{|2}$)

$$\begin{bmatrix}1 & -3 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 3 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}.$$

$\mathbf{I}_{\tau_{[(-\lambda)i+j]}}$ "undo" $\mathbf{I}_{\tau_{[(\lambda)i+j]}}$

How to undo $\mathbf{I}_{\tau_{[(\alpha)i]}}$?

$$\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 2\end{bmatrix}\begin{bmatrix}1 & & \\ & 1 & \\ & & 1/2\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}.$$





## 10 Interchange or swap matrices

Which matrix exchanges the columns?

$$\begin{bmatrix}a & c\\ b & d\end{bmatrix}\begin{bmatrix}\quad & \quad\\ & \\ & \end{bmatrix} = \begin{bmatrix}c & a\\ d & b\end{bmatrix}$$

Which matrix exchanges the rows? where do we put that matrix?

$$\begin{bmatrix}a & c\\ b & d\end{bmatrix}\qquad = \begin{bmatrix}b & d\\ a & c\end{bmatrix}$$

Matrix multiplication is not commutative!

## 11 Interchange of columns

### Interchange of columns:

$\mathbf{A}_{\underset{[i \rightleftharpoons j]}{\tau}} \rightarrow$ switch columns $i$ and $j$ of $\mathbf{A}$

$$\begin{bmatrix} 1 & -3 & 0 \\ 1 & -6 & 3 \end{bmatrix}_{\underset{[2 \rightleftharpoons 3]}{\tau}} = \begin{bmatrix} 1 & 0 & -3 \\ 1 & 3 & -6 \end{bmatrix}$$

We can switch two columns by a sequence of elementary transformations

Matrix $\mathbf{I}_{\underset{[i \rightleftharpoons j]}{\tau}}$ is called an exchange matrix





## 12 Permutation matrices

Product between exchange matrices $\mathbf{I}_{\underset{[\cdot \rightleftharpoons \cdot]}{\tau}}$ is a permutation matrix $\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}$.

$$\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}} = \text{Identity matrix } \mathbf{I} \text{ with rearranged columns}$$

Let's see the $3 \times 3$ case

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}, \quad \mathbf{I}_{\underset{[1 \rightleftharpoons 2]}{\tau}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

### How many $3 \times 3$ permutations can we find?

What happens if I multiply two permutation matrices?





## Questions of the Lecture 4

(L-4) Question 1.

(a) What are the matrices $\mathbf{I}_{\underset{[(x)1+2]}{\tau}}$, $\mathbf{I}_{\underset{[(y)1+3]}{\tau}}$ and $\mathbf{I}_{\underset{[(z)2+3]}{\tau}}$ that transform

$\mathbf{A} = \begin{bmatrix} 1 & 4 & -2 \\ 1 & 6 & 2 \\ 0 & 1 & 0 \end{bmatrix}$ into echelon form?

(b) Multiply those $\mathbf{I}_{\tau_i}$ to obtain a matrix $\mathbf{E}$ that performs the elimination: $\mathbf{AE} = \mathbf{K}$.

Based on (Strang, 1988, exercise 24 from section 1.4.)

(L-4) Question 2. Consider the matrix

$$\begin{bmatrix} 1 & 2 & 4 \\ -1 & -3 & -2 \\ 0 & 1 & c \end{bmatrix}$$

For what value(s) of $c$ is the matrix singular (meaning that we cannot find three pivots)?





(L-4) Question 3. Consider the following 3 by 3 matrices.

(a) $(\mathbf{I}_{\underset{[(-1)1+2]}{\tau}})$ subtracts column 1 from column 2 and then $(\mathbf{I}_{\underset{[2 \rightleftharpoons 3]}{\tau}})$ exchanges columns 2 and 3. What matrix $\mathbf{E}$ performs both changes at once?

(b) $(\mathbf{I}_{\underset{[2 \rightleftharpoons 3]}{\tau}})$ exchanges columns 2 and 3 and then $\mathbf{I}_{\underset{[(-1)1+3]}{\tau}}$ subtracts column 1 from column 3. What matrix $\mathbf{N} = (\mathbf{I}_{\underset{[2 \rightleftharpoons 3]}{\tau}})(\mathbf{I}_{\underset{[(-1)1+3]}{\tau}})$ performs both changes at once?

Explain why $\mathbf{E}$ and $\mathbf{N}$ are the same in both cases, while the elementary matrices $\mathbf{I}_{\tau}$ are different.

Based on (Strang, 1988, exercise 28 from section 1.4.)

(L-4) Question 4. The elementary matrices $\mathbf{I}_{\underset{[(?)1+2]}{\tau}}$ and $\mathbf{I}_{\underset{[(?)2+3]}{\tau}}$ reduce $\mathbf{A}$ to its echelon form. Find $\mathbf{E}$ so that $\mathbf{AE} = \mathbf{L}$ is an echelon form (lower triangular), if $\mathbf{A}$ is

$$\begin{bmatrix} 2 & 2 & 0 \\ 1 & 4 & 9 \\ 1 & 3 & 9 \end{bmatrix}$$

(L-4) Question 5. Although we will only consider the *Type I* and *II* elementary transformations here, most Linear Algebra textbooks include a third type: the *exchange* of columns

$$\mathbf{A}_{\underset{[p \rightleftharpoons s]}{\tau}} \rightarrow \text{Exchanges columns } p \text{ and } s \text{ of } \mathbf{A}.$$

Prove that a column exchange is, in fact, a sequence of *Type I* and *II* elementary transformations. Do this by transforming the identity matrix $\mathbf{I}_{2\times 2}$ into $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ using column elementary transformations.

(L-4) Question 6. Write the 3 by 3 matrices that produce the following elimination steps:

(a) $\mathbf{I}_{\underset{[(-5)\mathbf{1}+\mathbf{2}]}{\tau}}$ subtracts 5 times column 1 from column 2,

(b) $\mathbf{I}_{\underset{[(-7)\mathbf{2}+\mathbf{3}]}{\tau}}$ subtracts 7 times column 2 from column 3,

(c) $\mathbf{I}_{\underset{[\mathbf{G}]}{\tau}}$ exchanges columns 1 and 2, and then columns 2 and 3.

(Strang, 2003, exercise 1 from section 2.3.)





(L-4) Question 7. Consider the matrices from Problem 6:

(a) when multiplying by $\mathbf{I}_{\underset{[(-5)\mathbf{1}+\mathbf{2}]}{\tau}}$ and then by $\mathbf{I}_{\underset{[(-7)\mathbf{2}+\mathbf{3}]}{\tau}}$ the matrix $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$ we obtain $\mathbf{A}_{\underset{\substack{[(-5)\mathbf{1}+\mathbf{2}] \\ [(-7)\mathbf{2}+\mathbf{3}]}}{\tau}} = \begin{bmatrix} \; ; & ; & \; \end{bmatrix}$.

(b) But, when multiplying by $\mathbf{I}_{\underset{[(-5)\mathbf{1}+\mathbf{2}]}{\tau}}$ before and then by $\mathbf{I}_{\underset{[(-7)\mathbf{2}+\mathbf{3}]}{\tau}}$ we get $\mathbf{A}_{\underset{\substack{[(-7)\mathbf{2}+\mathbf{3}] \\ [(-5)\mathbf{1}+\mathbf{2}]}}{\tau}} = \begin{bmatrix} \; ; & ; & \; \end{bmatrix}$.

(c) When $\mathbf{I}_{\underset{[(-7)\mathbf{2}+\mathbf{3}]}{\tau}}$ comes first, the column ____ is not affected by column ____.
This fact is critical for how the LU factorization works!

(Strang, 2003, exercise 2 from section 2.3.)

(L-4) Question 8. What matrix $\mathbf{M}$ transforms the vector $\boldsymbol{v} = (1, \quad 0,)$ to $(0, \quad 1,)$, es decir $\boldsymbol{v}\mathbf{M} = (0, \quad 1,)$ ; and also transforms the vector $\boldsymbol{w} = (0, \quad 1,)$ to $(1, \quad 0,)$, es decir $\boldsymbol{w}\mathbf{M} = (1, \quad 0,)$?

(L-4) Question 9. Consider an interchange matrix $\mathbf{I}_{\underset{[i \rightleftharpoons j]}{\tau}}$, if we compute the product $\mathbf{A}(\mathbf{I}_{\underset{[i \rightleftharpoons j]}{\tau}})$, we get a new matrix like $\mathbf{A}$, but with exchanged columns. What happen if we compute the product $(\mathbf{I}_{\underset{[i \rightleftharpoons j]}{\tau}})\mathbf{A}$? Check your answer with a 2 by 2 example.





(L-4) Question 10. If each column of $\mathbf{A}$ is a multiple of $(1, \quad 1, \quad 1,)$, then $\mathbf{A}\boldsymbol{x}$ is always a multiple of $(1, \quad 1, \quad 1,)$. Write a 3 by 3 example. How many pivots are produced by elimination?
(Strang, 1988, exercise 26 from section 1.4.)





## 1 Highlights of **Lesson 5**

## Highlights of Lesson 5

- Inverse of $\mathbf{A}$
- Gauss-Jordan elimination / finding $\mathbf{A}^{-1}$
- Inverse of $\mathbf{AB}$, $\mathbf{A}^{\mathsf{T}}$

## 2 Inverse of a matrix (square matrices)

**A** squared of order $n$ has an inverse (is *invertible*) if it exists **B** such that

$$\mathbf{AB} = \mathbf{BA} = \mathbf{I}.$$

Then

$$\mathbf{B} = \mathbf{A}^{-1} \qquad \text{and} \qquad \mathbf{A} = \mathbf{B}^{-1}.$$

Not all matrices have inverses

*Square matrices with no inverses* are called *singular* matrices

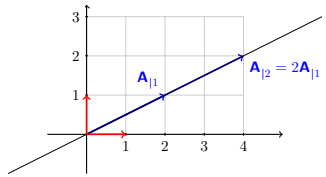




## 3 Singular case (no inverse)

$$\mathbf{A} = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

Is it possible to find a matrix **B** such that **AB** =**I**?
. . . columns of **I** must be linear combinations of columns of **A**. . . but the columns are aligned.




So

$$\boxed{\mathbf{A} \text{ is singular}}$$





## 4 Singular case (no inverse)

Can we find $\boldsymbol{x} \neq \mathbf{0}$ such that $\mathbf{A}\boldsymbol{x} = \mathbf{0}$?

$$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{pmatrix} \quad \\ \quad \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

If $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ and $\boldsymbol{x} \neq \mathbf{0}$ $\Rightarrow$ there is no $\mathbf{A}^{-1}$

$$\boxed{\text{The existence of } \mathbf{A}^{-1} \text{ leads to a contradiction}}$$

If $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ and $\boldsymbol{x} \neq \mathbf{0}$ $\Rightarrow$ $\mathbf{A}^{-1}\mathbf{A}\boldsymbol{x} = \mathbf{A}^{-1}\mathbf{0}$ $\Rightarrow$ $\boldsymbol{x} = \mathbf{0}$.

When $\mathbf{A}^{-1}$ does exist

the only solution to $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ is $\boldsymbol{x} = \mathbf{0}$.





## 5 Calculating the inverse matrix

$$\mathbf{A}(\mathbf{A}^{-1}) = \mathbf{I}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

So. . . we are solving $m$ systems (of $m$ equations each)

$$\begin{bmatrix} 1 & 3 \\ 2 & \end{bmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \begin{bmatrix} 1 & 3 \\ 2 & \end{bmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

## 6 Gauss-Jordan: solving two linear systems at once

### Gauss-Jordan elimination (obtaining a reduced echelon form $\mathbf{R}$)

apply elementary transformations until a echelon matrix with only zeros to the left of each pivot (and all pivots equal to 1) is achieved

Let's solve the linear systems

$$\begin{bmatrix}1 & 3\\ 2 & 7\end{bmatrix}\begin{pmatrix}a\\ b\end{pmatrix}=\begin{pmatrix}1\\ 0\end{pmatrix} \qquad \begin{bmatrix}1 & 3\\ 2 & 7\end{bmatrix}\begin{pmatrix}c\\ d\end{pmatrix}=\begin{pmatrix}0\\ 1\end{pmatrix}$$

applying Gauss-Jordan elimination on $\mathbf{A}$ stacked with $\mathbf{I}$

$$\begin{bmatrix}\mathbf{A}\\ \hline \mathbf{I}\end{bmatrix}=\left[\begin{array}{cc}1 & 3\\ 2 & 7\\ \hline 1 & 0\\ 0 & 1\end{array}\right] \rightarrow \qquad \rightarrow \qquad =$$

If $\mathbf{R}=\mathbf{I}$, we have found $\mathbf{A}^{-1}$





## 7 Gauss-Jordan: Why does it work?

$$\begin{bmatrix}\mathbf{A}\\ \hline \mathbf{I}\end{bmatrix}=\left[\begin{array}{cc}1 & 3\\ 2 & 7\\ \hline 1 & 0\\ 0 & 1\end{array}\right] \xrightarrow{\tau\,[(-3)\mathbf{1}+\mathbf{2}]} \qquad \xrightarrow{\tau\,[(-2)\mathbf{2}+\mathbf{1}]}$$

that is, since $\mathbf{A}_{\tau_1\cdots\tau_k}=\mathbf{A}\left(\mathbf{I}_{\tau_1\cdots\tau_k}\right)$:

$$\begin{bmatrix}\mathbf{A}\\ \hline \mathbf{I}\end{bmatrix}_{\tau_1\cdots\tau_k}=\begin{bmatrix}\mathbf{A}_{\tau_1\cdots\tau_k}\\ \hline \mathbf{I}_{\tau_1\cdots\tau_k}\end{bmatrix}=\begin{bmatrix}\mathbf{A}\left(\mathbf{I}_{\tau_1\cdots\tau_k}\right)\\ \hline \mathbf{I}_{\tau_1\cdots\tau_k}\end{bmatrix}=\begin{bmatrix}\mathbf{I}\\ \hline \mathbf{I}_{\tau_1\cdots\tau_k}\end{bmatrix},$$

who is $\mathbf{I}_{\tau_1\cdots\tau_k}$?

therefore $\mathbf{A}^{-1}=$





## 8 Inverse of a product

If $\mathbf{A}$ and $\mathbf{B}$ are invertible and of the same order, $(\mathbf{AB})$ is invertible.

How do we find the inverse of $\mathbf{AB}$? Let's try using $(\mathbf{B}^{-1}\mathbf{A}^{-1})$ :

$$\mathbf{AB}\left(\mathbf{B}^{-1}\mathbf{A}^{-1}\right)=$$

$$\left(\mathbf{B}^{-1}\mathbf{A}^{-1}\right)\mathbf{AB}=$$





## 9 Inverse of a transpose matrix

$$\mathbf{AA}^{-1}=\mathbf{I}$$

Let's take the transpose of both sides

$$\left((\mathbf{A}^{-1})^{\mathsf{T}}\right)\mathbf{A}^{\mathsf{T}}=\mathbf{I}$$

Therefore

the inverse of $\mathbf{A}^{\mathsf{T}}$ is

## 10 Interchange matrices and permutation matrices

Are interchange matrices $\mathbf{I}_{\underset{[i \rightleftarrows j]}{\tau}}$, invertible?

It is easy to demonstrate that

$$\left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}\right)^{\mathsf{T}}\left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}\right) = \mathbf{I} \qquad \Longrightarrow$$





## 11 Characterization of invertible matrices

Given $\mathbf{A}$ of order $n$, the following statements are equivalent

1. No zero columns in $\mathbf{A}_{\tau_1 \cdots \tau_p} = \mathbf{K}$ (pre-echelon matrix).
2. $\mathbf{A}$ is invertible.
3. $\mathbf{A}$ is a product of elementary matrices.

$$\mathbf{A}_{\tau_1 \cdots \tau_k} = \mathbf{A}\left(\mathbf{I}_{\tau_1 \cdots \tau_k}\right) = \mathbf{I} \qquad \Rightarrow \qquad \mathbf{A} = \left(\mathbf{I}_{\tau_1 \cdots \tau_k}\right)^{-1}$$

where

$$\left(\mathbf{I}_{\tau_1 \cdots \tau_k}\right)^{-1} = \left((\mathbf{I}_{\tau_1}) \cdots (\mathbf{I}_{\tau_k})\right)^{-1} = (\mathbf{I}_{\tau_k^{-1}}) \cdots (\mathbf{I}_{\tau_1^{-1}}) = \mathbf{I}_{\tau_k^{-1} \cdots \tau_1^{-1}}$$





## Questions of the Lecture 5

(L-5) Question 1. Use the Gauss-Jordan elimination method to invert

(a) $\mathbf{A}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

(b) $\mathbf{A}_2 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$.

(c) $\mathbf{A}_3 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.

(Strang, 1988, exercise 6 from section 1.6.)

(L-5) Question 2.

(a) If $\mathbf{A}$ is invertible and $\mathbf{AB} = \mathbf{AC}$, prove quickly that $\mathbf{B} = \mathbf{C}$.
(b) If $\mathbf{A} = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$, find an example with $\mathbf{AB} = \mathbf{AC}$, but $\mathbf{B} \neq \mathbf{C}$.

(Strang, 1988, exercise 4 from section 1.6.)

(L-5) Question 3. Use the Gauss-Jordan method to invert the generic matrix $2 \times 2$

$$\mathbf{M} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

The matrix is invertible (not singular) only when ...





(L-5) Question 4. Use the Gauss-Jordan method to invert the following matrices.

$$\mathbf{A} = \begin{bmatrix} 3 & 1 & 2 \\ -1 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}; \qquad \mathbf{B} = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 4 & -2 \\ 1 & 3 & 1 \end{bmatrix}$$

(L-5) Question 5. If the 3 by 3 matrix $\mathbf{A}$ has $\mathbf{A}_{|1} + \mathbf{A}_{|2} = \mathbf{A}_{|3}$, show that $\mathbf{A}$ is not invertible by two different methods:

(a) Find a nonzero solution $\boldsymbol{x}$ to $\mathbf{A}\boldsymbol{x} = \mathbf{0}$.
(b) Elimination keeps $column\,1 + column\,2 = column\,3$. Explain why there is no third pivot.

(Strang, 1988, exercise 26 from section 1.6.)

(L-5) Question 6. Find the inverses of

(a) $\mathbf{A}_1 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 3 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{bmatrix}$.

(b) $\mathbf{A}_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1/2 & 1 & 0 & 0 \\ 0 & -2/3 & 1 & 0 \\ 0 & 0 & -3/4 & 1 \end{bmatrix}$.

(c) $\mathbf{A}_3 = \begin{bmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{bmatrix}$.

(Strang, 1988, exercise 10 from section 1.6.)





(L-5) Question 7. Find the inverse of

$$\mathbf{A} = \begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix}$$

What values of $a$ and $b$ make the matrix singular?
(Strang, 1988, exercise 42 from section 1.6.)

(L-5) Question 8. Find $\mathbf{E}^2$, $\mathbf{E}^8$ and $\mathbf{E}^{-1}$ if $\mathbf{E} = \begin{bmatrix} 1 & 0 \\ 6 & 1 \end{bmatrix}$
(Strang, 1988, exercise 6 from section 1.5.)

(L-5) Question 9. Consider the following permutation matrix:

$$\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Find $\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}{}^{-1}$. Can you say something else about the relationship between $\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}$ and $\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}{}^{-1}$?





(L-5) Question 10. The 3 by 3 matrix $\mathbf{A}$ reduces to the identity matrix $\mathbf{I}$ by the following three column operations (in order):

$\underset{[(-4)\mathbf{1}+\mathbf{2}]}{\tau}$: Subtract 4 times column 1 from column 2.

$\underset{[(-3)\mathbf{1}+\mathbf{3}]}{\tau}$: Subtract 3 times column 1 from column 3.

$\underset{[(-1)\mathbf{3}+\mathbf{2}]}{\tau}$: Subtract column 3 from column 2.

(a) Write $\mathbf{A}^{-1}$ in terms of elementary matrices $\mathbf{I}_{\tau}$. **Then compute $\mathbf{A}^{-1}$.**
(b) What is the original matrix $\mathbf{A}$?

(Based on *MIT Course 18.06 Quiz 1, October 4, 2006*)





(L-5) Question 11. The 3 by 3 matrix $\mathbf{A}$ reduces to the identity matrix $\mathbf{I}$ by the following three **row** operations (in order):

$\underset{[(-4)\mathbf{1}+\mathbf{2}]}{\tau}$: Subtract 4 times row 1 from row 2.

$\underset{[(-3)\mathbf{1}+\mathbf{3}]}{\tau}$: Subtract 3 times row 1 from row 3.

$\underset{[(-1)\mathbf{3}+\mathbf{2}]}{\tau}$: Subtract row 3 from row 2.

(a) Write $\mathbf{A}^{-1}$ in terms of the $\mathbf{E}$'s. **Then compute $\mathbf{A}^{-1}$.**
(b) What is the original matrix $\mathbf{A}$ ?

(*MIT Course 18.06 Quiz 1, October 4, 2006*)

(L-5) Question 12.

(a) Find the inverse of $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

(b) Find the inverse of the following matrix **using the Gauss-Jordan method**

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ a & b & c & d \end{bmatrix}$$

(Poole, 2004, exercise 36, 38 and 59 from section 3.3.)

(L-5) Question 13. Consider the squared matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$. True or false?

(a) If $\mathbf{AB} = \mathbf{I}$ and $\mathbf{CA} = \mathbf{I}$ then $\mathbf{B} = \mathbf{C}$.

(b) $(\mathbf{AB})^2 = \mathbf{A}^2\mathbf{B}^2$.





(L-5) Question 14. Consider the matrix $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 2 \\ 1 & a & 0 & 2a \\ a & 0 & 1 & 0 \\ 1 & 0 & a & 1 \end{bmatrix}$

(a) Prove that $\mathbf{A}$ is invertible for any value of $a$.

(b) Compute $\mathbf{A}^{-1}$ when $a = 0$.

(L-5) Question 15. Consider the matrix $\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}$. Find $\mathbf{A}^{-1}$.

(L-5) Question 16. Find (if possible) the inverses of the following matrices

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}; \qquad \mathbf{B} = \begin{bmatrix} 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{bmatrix}.$$

(L-5) Question 17. There is a finite number ($n!$) of $n \times n$ permutation matrices. In addition, any power of a permutation matrix is also a permutation matrix. Use these facts to prove that $\left(\mathbf{I}_{\underset{[\mathfrak{S}]}{\tau}}\right)^r = \mathbf{I}$ for some integer $r$.





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